PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

CARD 07 / 10 • PROB-01

MATHEMATICAL REFERENCE

STUDIO CONTINUUM: 1·Ten-Frames | 2·Dominoes | 3·Clock | 4·Fractions | 5·Primes | 6·Algebra | **7·Probability** | 8·Volvelle | 9·Quest | 10·Rack | 11·SC-32

1. WHAT IS THIS MANIPULATIVE?

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.
Rolling two fair 6-sided dice gives 36 outcomes:
Sample Space = 6 × 6 = 36 Pair Outcomes
2d6 sum forms a triangle distribution peak at 7!

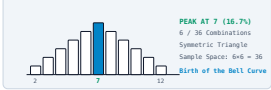
CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot
- Dials: 10-sector uniform & binary coin-flip

2d6 SAMPLE SPACE GRID & SPINNER POST

Lid features 6×6 outcome matrix & spinner post.
Watch experimental trials converge to exact odds!

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/36)



PEAK AT 7 (16.7%)
6 / 36 Combinations
Symmetric Triangle
Sample Space: 6×6 = 36
Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Precision Fair d6	4x	16.0 mm Volume-Balanced	High-Contrast
Polyhedral d20 Die	1x	20-Sided Icosahedron	Emerald Green
Low-Friction Spinner	1x	Brass Pivot Precision Hub	High-Spin Brass
Interchangeable Dials	4x	10-Sector / Binary / Custom	Polymer Plates
2d6 Sample Space Lid	1x	36-Outcome Matrix Grid	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. WHY KEEP THIS REFERENCE CARD?

✓ **NO LOST PIECE LEFT BEHIND**

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ **10-SECOND SELF-CHECKING PACK-UP**

Pack-up is a built-in math game! Seat the 4 d6 dice and 1 d20 into their molded bottom bays. Stack the spinner dials flat and click the brass spinner hub onto its retention post. Any empty bay shows. Lid latches firmly for silent, rattle-free carry.

✓ **HANDS-ON TUTORIAL ON REVERSE**

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

CENTRAL LIMIT THEOREM & LAW OF LARGE NUMBERS
The sum of independent random variables tends toward a normal distribution as sample size grows.
Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL
stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 07 OF 10 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS • 2d6 36-OUTCOME MATRIX • WEIGHTED HARMONIC SPINNER

LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

Grades 5–10 Probability & Statistics • Hands-On Classroom & Home STEM Studio Series

CARD 07 / 10 • PROB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN
stackcalc.io/products/dice-and-spinner/
Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.7.SP.C.8 (Compound Events)
- CCSS.MATH.CONTENT.HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representations)

Grades 5–10 Probability & Statistics

MATHEMATICAL INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

1 STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1–6) × Die 2 columns (1–6) = 36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

2 STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:
- Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
- Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P = 6/36 (16.7%)
- Sum 12: Only 1 way [6,6] → P = 1/36 (2.8%)

3-4 STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.
4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:
Press Dice & Statistics Icon ([▚▞], [Σ+] & [g][X]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.
Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60.

5 CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together, extreme sums require extreme coordination, while central sums have many different additive combinations! This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)
→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

STACKCALC ARCHIVAL INSTRUMENT LABS
Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION
Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL
stackcalc.io/parts
Classroom Lab Crates: stackcalc.io/schools

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR SNAPFINITY STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PROB-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] SNAPFINITY ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 14 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California